

Bayesian logistic regression

Put a Gaussian prior on the weights, keep the whole posterior, and predict by averaging the sigmoid over it.

THE QUESTION

Logistic regression has no conjugate prior. How do we get its posterior, and what does it change?

THE INTUITION

Multiply a Gaussian prior by the Bernoulli likelihood. The product has no closed-form normaliser, so use a grid (two weights) or samples (any number). Average the samples' sigmoids to predict.

KEY TAKEAWAY

Predict with $\frac{1}{M} \sum_m \sigma(\theta_m^T x^*)$ over posterior samples, not the MAP's sigmoid. They agree near the data; far from it the average moves to 0.5.

Bernoulli likelihood → × Gaussian prior → no closed form → grid · Laplace · MCMC → average $\sigma(\theta^T x^*)$ → $p(y^* = 1 | x^*, D)$

1 A sigmoid of $\theta^T x$, with no bias term

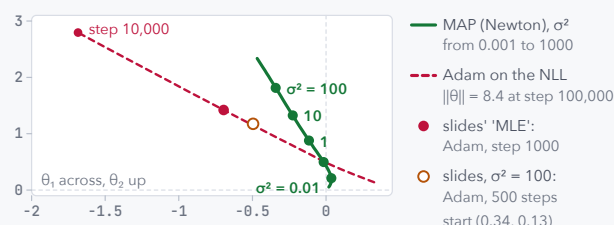


Each label is a coin, $y_i \sim \text{Bernoulli}(\sigma_i)$ with $\sigma_i = \sigma(\theta^T x_i)$:

$$\log p(y | X, \theta) = \sum_i [y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)].$$

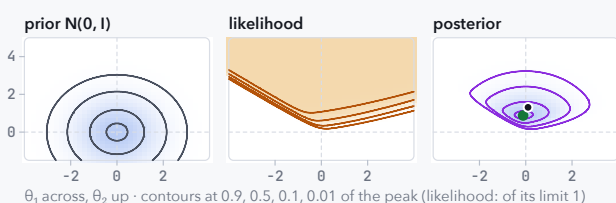
Prior $\theta \sim \mathcal{N}(0, \sigma^2 I)$; this σ^2 is a variance, not the sigmoid. With no bias every boundary $\theta^T x = 0$ passes through the origin.

2 Separable data: the MLE runs off, the prior stops it



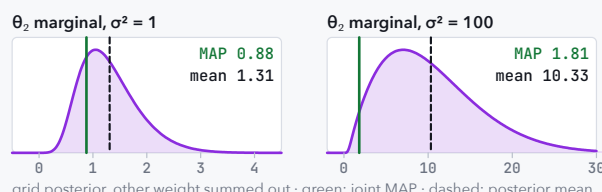
The gradient $X^T(\sigma(X\theta) - y)$ has no closed-form root. $\theta = (1, 1)$ already separates the classes, so scaling θ up lowers the NLL for ever. The MAP minimises $f(\theta) = -\log p(D | \theta) + \theta^T \theta / (2\sigma^2) + \log 2\pi\sigma^2$ and is finite: at $\sigma^2 = 100$ it is $(-0.342, 1.815)$, while the slides' $(-0.496, 1.175)$ is Adam after 500 steps ($f = 6.505$, not 6.465).

3 Prior times likelihood, normalised on a grid



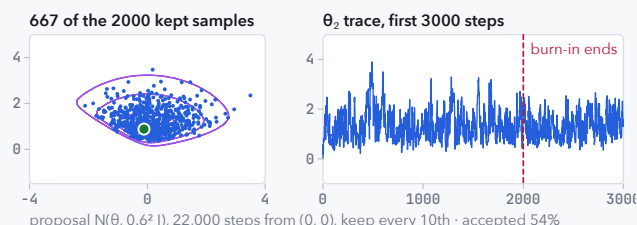
$p(\theta | D) = p(D | \theta)p(\theta)/p(D)$ with $p(D) = \int p(D | \theta)p(\theta)d\theta$. Not conjugate, so no closed form; with two weights a 161×131 grid sums it ($\log p(D) = -1.58$ at $\sigma^2 = 1$). The likelihood is a wedge near 1 out to infinity; the prior closes it.

4 The posterior leans outward: mean and mode differ



The likelihood is flat as $\|\theta\|$ grows and steep the other way. At $\sigma^2 = 1$: MAP $\theta_2 = 0.88$, mean 1.31, 90% interval 0.58 to 2.32; at $\sigma^2 = 100$: MAP 1.81, mean 10.3. $P(\theta_1 > 0) = 0.53$ though the MAP's θ_1 is -0.12 . A Gaussian at the MAP (the Laplace approximation, next sheet) cannot lean.

5 Random-walk Metropolis draws the samples



Propose $\theta' = \theta + \epsilon$, $\epsilon \sim \mathcal{N}(0, 0.6^2 I)$; accept with probability $\min(1, \tilde{p}(\theta')/\tilde{p}(\theta))$, $\tilde{p}$ = likelihood × prior, so the evidence cancels. From (0, 0), 22,000 steps, 54% accepted; drop the first 2000 (burn-in) and keep every 10th: 2000 samples. Mean (0.16, 1.29), sd (0.80, 0.53); grid (0.10, 1.31), (0.81, 0.54). More in the MCMC sheet.

6 Three ways round the missing normaliser

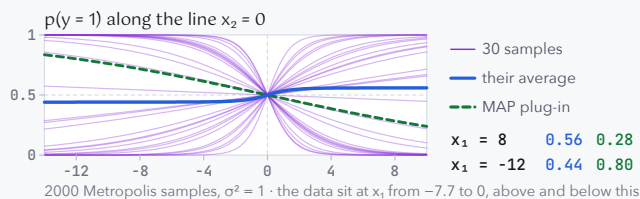
route	gives	catch
MAP only	one point θ_{MAP}	no uncertainty
grid	exact, to grid resolution	G^D points
Laplace	$\mathcal{N}(\theta_{\text{MAP}}, H^{-1})$, $H = \nabla^2 f$	symmetric, one mode
MCMC	correlated samples	tuning, burn-in

All three need only $p(D | \theta)p(\theta)$. The slides take the Laplace route (next sheet) and average over 1000 of its samples.

Predict by averaging over weights

Every posterior sample is a boundary. Their average sigmoid is the prediction: sure near the data, hedging away from it.

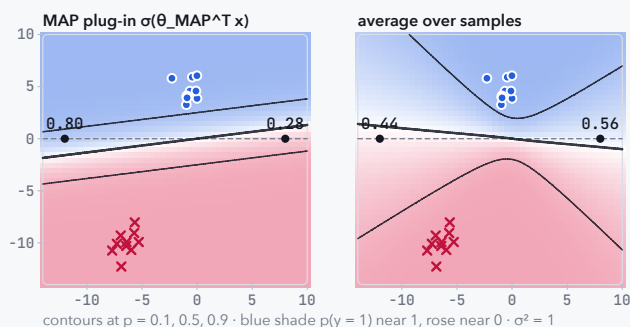
7 The predictive averages sigmoids



$$p(y^* = 1 \mid x^*, \mathcal{D}) = \int \sigma(\theta^T x^*) p(\theta \mid \mathcal{D}) d\theta \approx \frac{1}{M} \sum_{m=1}^M \sigma(\theta_m^T x^*)$$

On $x_2 = 0$ the sampled sigmoids are steep and disagree in direction, so their mean stays near 0.5; the MAP's claims 0.80 at $x_1 = -12$. Average probabilities, not weights: $\mathbb{E}[\sigma(\theta^T x)] \neq \sigma(\mathbb{E}[\theta]^T x)$.

8 Near the data they agree; away from it the average drifts to 0.5



Between the clusters, at $(-3, -3)$: 0.08 (average) and 0.09 (plug-in). Beyond the blue cluster, $(-8, 8)$: 0.88 and 1.00; at $(4, -6)$: 0.06 and 0.003. The plug-in stays sure wherever $\theta_{\text{MAP}}^T x$ is large, data or not.

9 Pause and think

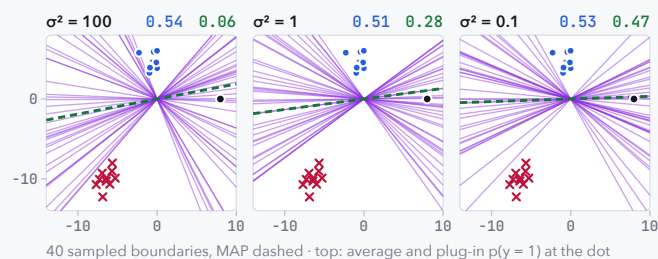
Separable data: does the MAP exist? Yes, for any finite σ^2 : the penalty grows while the NLL only falls towards 0.

Why is the posterior mean further out than the MAP? The likelihood is flat outward and steep inward.

Where do predictive and plug-in differ most? Far from the data, where sampled boundaries disagree.

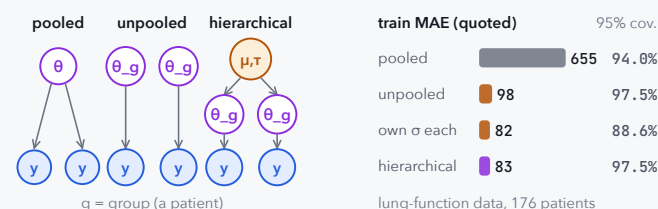
How many samples? The error is $\text{sd} / \sqrt{M}$: 0.01 at $M = 2000$, more for correlated draws (grid 0.53 at $(8, 0)$, samples 0.56).

10 Boundary samples fan out; the prior sets the spread



Each boundary is a line through the origin; the fan shows the angle's uncertainty (90% within -54° to 43° at $\sigma^2 = 1$). Weak prior: long weights, near-step sigmoids, yet the average at $(8, 0)$ is 0.54 against the plug-in's 0.06. Strong prior: even the plug-in is soft (0.47).

11 Hierarchical priors pool across groups



Hierarchical: $\theta_g \sim \mathcal{N}(\mu, \tau^2)$ with priors on μ and τ , so a group with few points is pulled towards μ . The notebook fits linear regressions (NUTS, 1549 visits); a logistic model takes the same prior. Training MAE favours unpooled fits; pooling helps sparse or new groups.

12 Metropolis and the predictive in NumPy

```
sig = lambda z: 1 / (1 + np.exp(-z))
def log_post(th, X, y, s2): # unnormalised
    z = X @ th
    ll = -(np.logaddexp(0, z) - y*z).sum()
    return ll - th @ th / (2*s2)
rng = np.random.default_rng(0)
th, S = np.zeros(2), []
lp = log_post(th, X, y, 1.0)
for _ in range(22000):
    prop = th + 0.6 * rng.normal(size=2)
    lq = log_post(prop, X, y, 1.0)
    if np.log(rng.random()) < lq - lp:
        th, lp = prop, lq
    S.append(th)
S = np.array(S)[2000::10] # 2000 kept
p = sig(X_new @ S.T).mean(1) # predictive
```

no closed-form posterior $\rightarrow$ grid, Laplace or MCMC $\rightarrow$ average $\sigma(\theta_m^T x^*)$ over samples · sure near the data, near 0.5 far away

Source: lecture slides "Bayesian logistic regression" and the notebooks on Bayesian logistic regression and hierarchical regression (outputs quoted), nipunbatra.github.io/pml-teaching. Data reproduced exactly; the Adam runs, replayed in NumPy, give the slides' weights; grid, Newton and Metropolis (fixed seeds) go beyond the slides. Corrections: the weak-prior "MAP" is unconverged (panel 2); the notebook's Laplace covariance at $\sigma^2 = 0.01$ counts the likelihood 20 times (broadcasting; diagonal 0.0020, 0.0006, correct 0.0079, 0.0048); the two-weight log prior constant is $-\log 2\pi\sigma^2$, not $-\frac{1}{2} \log 2\pi\sigma^2$.